CO1 - AT1

Shaik mohammad Athim
192525132
CSA 1513

1. **Defination:** Cloud computing is the delivery of computing an resources such as servers, storage, database, networking and software over the internet. users can access these resources whenever reauired without owning and maintaining physical infrastructure

Four characteristics:

1. on-demand self service: users can create or use computing resources whenever needed without human interaction from the services provider.
2. Broad network access : cloud services can be accessed through the internet using computers, materials, and other devices.
3. Resource Pooling: Cloud providers share Physical resources among many users efficiently.
4. Rapid elasticity: Resources can be increased or decreased quickly according to demand

Difference from traditional computing: In additional on-premise system reauire organizations to purchase, install and maintain their own hardware and software, while cloud computing provids resources Through the internet

2. Cloud computing developed gradually Through several stages:
   1. mainframe computing: Large maintance computers were shared by many users Through terminals.
   2. personal computers: Application and data were distributed b/w client computer and central server.

3. Client-server computing:- Applications and data were distributed b/w client computers and central servers.

4. internet development:- The growth of the internet allowed users to acces applications and information from anywhere.

5. Virtualization:- one physical server could be divided into multiple virtual machines. This improved hardware utilization and reduced costs.

6. web services and SOA:- Software components could communicate over network and be provided as services.

mainframes → PCs → client servers → internet →

Virtualization → webservices → Cloud computing

3. Role in cloud computing

* improve hardware utilization
* Allows multiple users to share physical resources.
* Reduces hardware costs
* provides isolation b/w users
* make resource allocation and scaling easier
* Allow virtual machines to be created and removed quickly.

Difference:

Virtualization

* it is a technology
* creates virtual resources from physical resources
* concerns without cloud computing

Cloud computing

* it is a service delivery model
* provides computing resources to users to over a network.
* commonly uses virtualization as an enabling technology

4. Roles in data centers:

1. Computing:- Host virtual machines and applications
2. Storage:- store user files, database & application data.
3. Networking:- Connect cloud resources to users through the internet

4. Security :- use firewalls access controls and maintaing to protect data.

5 Availability :- use backup systems and redundant equipment to keep services running.

6. Scalability :- Add more servers and resources when demand increases.

7 Disaster recovery :- maintains backups and alternative systems to reduce data loss.

5. Iaas, Paas, saas and Xaas

model	memory	Example
Iaas	infrastructure as a service	Amazon EC2
Paas	Platform as a service	google APP Engine
Saas	Software as a service	google APP Engine
Xaas	Anything / Everything as a service	Different IT services delivered by cloud

Iaas - virtual servers, Storage networks

Paas - Development platform and tools

Saas - Ready to use software

Xaas - Different IT services delivered through cloud.

12/10/26